Deeksha Dangwal

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Research Interests

AI Safety and Alignment, ML Systems, Security and Privacy
I am a systems researcher working on AI safety. My background is in computer architecture and hardware security, and I study where safety mechanisms should be enforced in the training stack. Prior work has shown that safety training in open-weight models can be undone with a small amount of fine-tuning; my current work develops a threat model for this setting and monitors that operate on model weights during fine-tuning. My work in computer architecture, privacy, and secure computation appeared in ISCA, ASPLOS, PLDI, FPL, BMVC, and TMLR. My SoK on hardware-software-security codesign was **cited in the US National Strategy on Microelectronics Research**, I am a **Rising Stars in EECS** alumna, and trace wringing was an **IEEE Micro Top Pick**.

Education

Doctor of Philosophy, Computer Science **03/2022**
University of California, Santa Barbara
Thesis: **"A System-level Framework for Privacy"** advised by **Prof. Timothy Sherwood**

Master of Science, Electrical and Computer Engineering **09/2016**
University of California, Santa Barbara

Bachelor of Engineering, Instrumentation and Electronics **05/2014**
Ramaiah Institute of Technology, Bangalore, India

Awards and Honors

2026 ERA:AI Fellowship in Technical AI Governance, Cambridge, UK *(Top 0.5% of applicants)* **2026**

Our paper, "SoK: Opportunities for Software-Hardware-Security Codesign for Next-Generation Secure Computing" (HASP'20) appears in the *United States National Strategy on Microelectronics Research* **2024**

Runner-up, UCSB Grad Slam Competition **2021**

Rising Stars in EECS, UC Berkeley **2020**

IEEE Micro Top Pick, Trace Wringing for Program Trace Privacy **2020**

Fiona and Michael Goodchild Graduate Mentoring Award, UCSB Graduate Division **2020**

Outstanding Graduate Student Award, Department of Computer Science, UCSB **2020**

Experience

Fellow, *MATS Research, Berkeley, CA* **09/2026 - Present**

- Compute Verification: Designing and empirically evaluating software-based compute verification protocols (memory-state challenges, subsampled zero-knowledge proofs) that bound undeclared GPU workloads without trusted hardware *(Mentored by James Petrie, Future of Life Institute)*

Research Fellow, *ERA:AI Fellowship in Technical AI Governance, Cambridge, UK* **07/2026 - Present**

- Training-Time Safety Monitoring: Developed a defense against fine-tuning attacks that strip safety behavior from open-weight LLMs. Built a testbed of over 200 controlled fine-tuning runs spanning four open reasoning model families, from 20B to 1T parameters, and designed a monitor that watches only the weight updates as training runs. The monitor catches safety-removal with AUROC 0.97 to 1.00 before the model's behavior collapses and transfers across model families and training objectives with no per-model tuning. (A3). *(Mentored by Alan Cooney, Google DeepMind)*

Consultant, *Office of the CTO, The Allen Institute, Seattle, WA* **07/2025 - 09/2025**

- Neural tracing: Co-developed neural tracing for mechanistic interpretability of LLMs and bio-foundation models. Submitted patent application *"Model Optimization and Data Analysis Using Neural Traces"* (P2)

Research Scientist, *Reality Labs Research, Meta* **01/2022 - 04/2025**

- Distributed Intelligence and On-device AI: Prototyped gesture interaction for smart glasses (data collection, model training, compression, and demo) on SoC with custom and off-the-shelf accelerators. Through HW/SW codesign and energy-aware scheduling, my pipeline improved latency and power (by 93% & 74%)
- Performance and Power Modeling: Built Python-based system-level power and performance modeling tool for Project Aria and other wearable devices using discrete-event simulation. The system latency and power was calibrated against measurements on prototype devices running machine perception pipelines: eye tracking, hand and gesture tracking, localization and mapping.
- Hardware-Software-Security Codesign in Extended Reality: Developed security and privacy research program for mixed and augmented reality, establishing threat models for codec avatars, privately crowdsourced data collection for world modeling, and other extended reality applications; published in HASP 2020 (W3), PLDI 2021 (C5), BMVC 2021 (C4), TMLR 2025 (J4, A1)

Graduate Student Researcher, *ArchLab, UC Santa Barbara* **06/2015 - 12/2021**

- Trace Wringing: Developed trace wringing for safer program behavior sharing, achieving privacy through lossy compression and information flow tracking with verifiable leakage bounds; published in ASPLOS 2019, SEED 2021 (C3, C6) and recognized as IEEE Micro Top Pick (J2)
- Agile Hardware Development with PyRTL: Co-developed PyRTL, a Pythonic hardware development toolkit enabling rapid prototyping and agile hardware design; published in FPL 2017 (C1) and IEEE Micro 2020 (J3)
- Architecture Modeling with Charm: Worked on Charm, a domain-specific language for high-level architecture modeling; published in ISCA 2018, JETC 2019 (C2, J1)
- OpenTPU: Led development in PyRTL, creating open-source tensor processing unit implementation

Research Scientist Intern, *Reality Labs Research, Meta* **06/2020 - 01/2021**

- Implemented novel reverse engineering attack on local feature descriptors, surpassing state-of-the-art reconstruction accuracy for user image recovery
- Established first privacy threat model for computer vision feature descriptor sharing in AR systems
- Developed privacy-preserving mitigation techniques and studied effects on downstream vision system performance; published in BMVC 2021 (C4)

Research Intern, *Microsoft Research* **06/2018 - 09/2018**

- Implemented parameterizable architecture-aware machine learning graph primitives for custom hardware instructions on Brainwave Neural Processing Unit
- Built tools for automatic lifting of hardware instructions to high-level graph primitives while maintaining hardware fidelity
- Designed computational pipeline for NPU model decompilation with accuracy verification; resulted in patent (P1)

Research Assistant, *Oracle Labs* **06/2016 - 09/2016**

- Established testing environment for measuring throughput of RAPID Data Processing Unit (DPU) network, a bandwidth-optimized big data computation architecture
- Implemented network congestion tests for best/worst case traffic using hardware RPC acceleration mechanisms

Publications

[C]=Conference, [J]=Journal or Magazine, [W]=Workshop

W4 **The Need for Computational Pluralism**
D. Dangwal, A. Rajagopal. *Workshop on Ethical Systems and Architecture Design **(HotEthics)**, 2026*

J4 **Unlocking Visual Secrets: Inverting Features with Diffusion Priors for Image Reconstruction**
S. Q. Zhang, Z. Li, C. Guo, S. Mahloujifar, D. Dangwal, E. Suh, B. D. Salvo, C. Liu. *Transactions on Machine Learning Research **(TMLR)**, 2025*

C6 **Context-Aware Privacy-Optimizing Address Tracing**
D. Dangwal, Z. Zhang, J. Crandall, T. Sherwood. *IEEE International Symposium on Secure and Private Execution Environment Design **(SEED)**, 2021*

C5 **Porcupine: A Synthesizing Compiler for Vectorized Homomorphic Encryption**
M. Cowan, D. Dangwal, A. Alaghi, C. Trippel, V. T. Lee, B. Reagen. *Programming Language Design and Implementation **(PLDI)**, 2021*

C4 **Mitigating Reverse Engineering Attacks on Local Feature Descriptors**
D. Dangwal, V. T. Lee, H. J. Kim, T. Shen, M. Cowan, R. Shah, C. Trippel, B. Reagen, T. Sherwood, V. Balntas, A. Alaghi, E. Ilg. *British Machine Vision Conference **(BMVC)**, 2021*

W3 **SoK: Opportunities for Software-Hardware-Security Codesign for Next Generation Secure Computing**
D. Dangwal, M. Cowan, A. Alaghi, V. Lee, B. Reagen, C. Trippel. *Hardware and Architectural Support for Security and Privacy **(HASP)**, 2020*

J3 **Agile Hardware Development and Instrumentation with PyRTL**
D. Dangwal, G. Tzimpragos, T. Sherwood. *IEEE Micro Special Topics on Agile & Open Source Hardware, 2020*

J2 **Trace Wringing for Program Trace Privacy**
D. Dangwal, W. Cui, J. McMahan, T. Sherwood. *IEEE Micro's Top Picks from Computer Architecture Conferences, 2020 **(IEEE Micro Top Pick)***

C3 **Safer Program Behavior Sharing through Trace Wringing**
D. Dangwal, W. Cui, J. McMahan, T. Sherwood. *Architectural Support for Programming Languages and Operating Systems **(ASPLOS)**, 2019*

J1 **Language Support for Navigating Architecture Design in Closed Form**
W. Cui, G. Tzimpragos, Y. Tao, J. McMahan, D. Dangwal, N. Tsiskaridze, G. Michelogiannakis, D. Vasudevan, T. Sherwood. *ACM Journal on Emerging Technologies in Computing Systems **(JETC)**, 2019*

W2 **PyRTLMatrix: an Object-Oriented Hardware Design Pattern for Prototyping ML Accelerators**
D. Aboye, D. Kupsh, M. Lim, J. Mai, D. Dangwal, D. Mirza, T. Sherwood. *Workshop on Energy Efficient Machine Learning and Cognitive Computing for Embedded Applications **(EMC2)**, 2019*

W1 **PyRTL in Early Undergraduate Research**
D. Mirza, D. Dangwal, T. Sherwood. *Workshop on Computer Architecture Education **(WCAE)**, 2019*

C2 **Charm: A Language for Closed-form High-level Architecture Modeling**
W. Cui, Y. Ding, D. Dangwal, A. Holmes, J. McMahan, A. JavadiAbhari, G. Tzimpragos, F. Chong, T. Sherwood. *International Symposium on Computer Architecture **(ISCA)**, 2018*

C1 **A Pythonic Approach for Rapid Hardware Prototyping and Instrumentation**
J. Clow, G. Tzimpragos, D. Dangwal, S. Guo, J. McMahan, T. Sherwood. *International Conference on Field-Programmable Logic and Applications **(FPL)**, 2017*

Pre-Prints, Reports, Articles

A3 **Position: Safety Monitoring Should Operate on Weights During Fine-Tuning**
D. Dangwal et al. *In preparation, 2026*

A2 **Mechanism Design for Improving Hardware Security Workshop Report** *CCC Workshop Report, August 2022*

A1 **Analysis and Mitigations of Reverse Engineering Attacks on Local Feature Descriptors**
D. Dangwal, V. T. Lee, H. J. Kim, T. Shen, M. Cowan, R. Shah, C. Trippel, B. Reagen, T. Sherwood, V. Balntas, A. Alaghi, E. Ilg. *arXiv preprint, May 2021*

Patents

P2 **Model Optimization and Data Analysis Using Neural Traces**
A. Rajagopal, G. H. Huynh, D. Dangwal *Provisional Patent Filed September 2025*

P1 **Deriving a concordant software neural network layer from a quantized firmware neural network layer**
J. Fowers, D. Lo, D. Dangwal *US Patent 11556764B2, Microsoft Technology Licensing LLC, 2023*

Invited Talks and Seminars

T8 **Performance, Power, and Privacy: Codesign Strategies for Always-On Wearables**
Washington State University, Pullman 06/2025

T7 **Privacy and Security in the age of Metaverse: A case for data minimization and on-device AI**
Washington State University, Everett 05/2025

T6 **A System-Level Framework for Privacy: Applications of Wringing in AR/VR**
Meta, Reality Labs Research 03/2021

T5 **A System-Level Framework for Privacy**
Arizona State University, Phoenix 03/2021

T4 **A System-Level Framework for Privacy**
Pennsylvania State University, State College 03/2021

T3 **Privacy through Wringing**
UCSB Grad Slam Final 03/2021

T2 **Agile hardware development and instrumentation with PyRTL**
Agile-RTL Workshop, UC Santa Barbara, CA 09/2019

T1 **OpenTPU: Prototyping ML Accelerators with PyRTL**
IEEE Space Computing Conference, Caltech, Pasadena, CA 09/2019

Teaching and Mentorship

Mentor, SPAR (Supervised Program for Alignment Research) **Fall 2026**

- Project: *Constraint Drift Through Delegation Hierarchies*, co-mentored with Abhejit Rajagopal (Allen Institute, UCSF)

Teaching Assistant, CS 154: Computer Architecture, UCSB Spring 2021

Teaching Assistant, Research Mentorship Program (RMP), UCSB Summer 2019

NSF ERSP Mentor, Department of Computer Science, UCSB 2018 - 2019

- Students published "PyRTLMatrix: an Object-Oriented Hardware Design Pattern for Prototyping ML Accelerators" at EMC2 workshop (W2)
- Awarded university-wide Fiona and Michael Goodchild Graduate Mentoring Award for excellence in mentorship

Women in STEM Mentorship Program, UCSB 10/2016 - 03/2018

Teaching Assistant, Department of Physics, UCSB 01/2015 - 06/2015

EUREKA Mentorship Program, California NanoSystems Institute, UCSB 06/2015 - 08/2015

Students Mentored

- Joann Chen (Research Intern, Meta, 2022), Manu Kondapaneni (2020), Junayed Naushad (2019), Dawit Aboye (2018-2019), Dylan Kupsh (2018-2019), Maggi Lim (2018-2019), Jacqueline Mai (2018-2019), Angela Yung (2016-2017), Saurabh Gupta (2015)

Professional Service and Activities

CRA Congressional Visit Day, Washington State Contingent 09/2025

- Represented CRA and Washington State in Washington, D.C., and joined meetings with Members of Congress to make the case for federal support of computing research

Invited Participant, CCC Future of AI Research in Industry 07/2025

Invited Participant, CCC Computing Futures Symposium 05/2025

Invited Participant, CCC Mechanism Design for Improving Hardware Security 08/2022

Organizer, Undergrad Architecture Mentoring Workshop (uArch) 2022 - 2025

Program Committee Member

- MICRO '26, ISPASS '26, HPCA ('26, '25, '24), ASPLOS '25, ISCA '24, SEED '24, HASP '23

External Review Committee

- MICRO '24, YArch ('25, '23), ISCA '22, SEED '21, ASPLOS '20 (Artifact Evaluation)

Co-President, Women in Computer Science (WiCS), UCSB 2018 - 2020

Graduate Representative for Faculty Recruitment, Department of Computer Science, UCSB 2019 - 2020

Graduate Representative for Diversity, Department of Computer Science, UCSB 2018 - 2019

Grace Hopper Celebration of Women 2017, 2018, 2019

CRA-W Grad Cohort 2017, 2018

(Updated October 2026)